CO2 Assessment Tool 2
Dry Run Evaluation

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1. Breadth-First Search (BFS)

Given: Start node A. The graph has edges A–B, A–C, B–D, B–E, C–F and C–G.

Algorithm: BFS explores nodes level by level using a FIFO queue. Visited nodes are not added again.

Iteration	Queue before expansion	Visited node	Queue after expansion
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

1. BFS Traversal Order: A → B → C → D → E → F → G.

2. Queue Contents: The queue changes after each expansion as shown in the table. BFS visits all nodes at one level before moving to the next.

2. Depth-First Search (DFS)

Given: The same graph, starting at A.

Algorithm: DFS explores one branch as deeply as possible before backtracking. Assuming left-to-right traversal, it visits the left subtree first.

Iteration	Current node	Stack (top at right)
1	A	C, B
2	B	C, E, D
3	D	C, E
4	E	C
5	C	G, F
6	F	G
7	G	Empty

DFS Traversal Order: A → B → D → E → C → F → G.

Explanation: DFS uses a stack (LIFO) or recursion. After reaching D, it backtracks to B and visits E, then returns to A to explore C's subtree.

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3. Eight-Puzzle Problem Using BFS

Initial State:

1	**3**	**6**
5	0	2
4	7	8

Goal State:

1	**2**	**3**
4	5	6
7	8	0

Here, 0 represents the blank tile. Each move slides an adjacent tile into the blank space. BFS explores states level by level, using a FIFO queue and avoiding previously visited states.

First five expansion iterations: The queue contains states waiting to be expanded. For compactness, each state is written row-wise; for example, `136/502/478` represents the three rows of the initial state.

Iteration	Current state expanded	Queue before (excluding current)	Queue after
1	136/502/478	Empty	106/532/478; 136/052/478; 136/520/478; 136/572/408

2	106/532/478	3 states remain	Previous 3 + 016/532/478 + 160/532/478
3	136/052/478	4 states remain	Previous 4 + 036/152/478 + 136/452/078
4	136/520/478	5 states remain	Previous 5 + 130/526/478 + 136/528/470
5	136/572/408	6 states remain	Previous 6 + 136/572/048 + 136/572/480

Note: The queue entries are abbreviated to show the newly generated states; previously queued states remain in their original FIFO order.

Eight-Puzzle: Questions and Answers

1. Which node is expanded in each iteration?

BFS expands the oldest state in the queue. The first five expanded states are:

1. `136/502/478`
2. `106/532/478`
3. `136/052/478`
4. `136/520/478`
5. `136/572/408`

The search continues until the goal state is reached.

2. How many states are generated in total?

The exact count depends on when the search stops and how duplicate states are handled. With duplicate-state checking and the neighbour

order used in this dry run, **456 distinct states are discovered by the time the goal state is removed from the queue**, and 283 states, including the goal, are expanded. The solution path itself contains 9 states.

3. Complete solution path from initial state to goal state

The shortest path contains 8 moves:

Step	State
0	136/502/478
1	136/520/478
2	130/526/478
3	103/526/478
4	123/506/478
5	123/056/478
6	123/456/078
7	123/456/708
8	123/456/780

4. Why does BFS find the shortest solution in an unweighted 8-puzzle?

BFS explores all states at depth 0, then depth 1, depth 2, and so on. Since every move has equal cost, the first goal state reached has the minimum number of moves. Therefore, BFS guarantees a shortest solution when duplicate states are handled correctly.